Preconditioned Primal-Dual Gradient Method for Nonconvex Nonlinear Composite Optimization

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1 Algorithm

When updating x, linearize the constraint $g(x)$ at x^k; when updating y, linearize the constraint $g(x)$ at x^{k+1}. Extrapolation is performed on the primal variable:

$$\begin{aligned}
x^{k+1} &= \arg\min_x \left\{ \left\langle \nabla f(x^k), x - x^k \right\rangle + \left\langle y^k, g(x^k) + J(x^k)(x - x^k) \right\rangle + \frac{1}{2\tau}\|x - x^k\|^2 \right\} \\
&= \arg\min_x \left\{ \left\langle \nabla f(x^k) + J(x^k)^T y^k, x - x^k \right\rangle + \frac{1}{2\tau}\|x - x^k\|^2 \right\}, \\
\widetilde{x}^{k+1} &= 2x^{k+1} - x^k, \\
y^{k+1} &= \arg\min_y \left\{ h^*(y) - \left\langle g(x^{k+1}) + J(x^{k+1})(\widetilde{x}^{k+1} - x^{k+1}), y \right\rangle + \frac{1}{2}\|y - y^k\|^2_{M_k} \right\} \\
&= \arg\min_y \left\{ h^*(y) - \left\langle g^{k+1}(\widetilde{x}^{k+1}), y \right\rangle + \frac{1}{2}\|y - y^k\|^2_{M_k} \right\},
\end{aligned}$$

where

$$M_k = \tau J(x^k) J(x^k)^T.$$

1.1 Preliminaries

Notations

$$\|A\| = \|A\|_F, \qquad \|a\| = \|a\|_2.$$

Some useful properties

1. For infinitely continuously differentiable functions, for every $i = 1, 2, \ldots, m$, $g_i(x)$ is $L^0_{g_i}$-Lipschitz continuous:
$$\|\nabla g_i(x)\| \le L^0_{g_i};$$
the gradient $\nabla g_i(x)$ is L_{g_i}-Lipschitz continuous:
$$H_{g_i}(x) \preceq L_{g_i} I, \qquad \lambda_{\max}(H_{g_i}(x)) \le L_{g_i}, \qquad \forall x \in \mathcal{X}.$$

2. The nonlinear mapping $g(x) : \mathbb{R}^n \to \mathbb{R}^m = (g_1(x), \ldots, g_m(x))^T$ is L^0_g-Lipschitz continuous, where
$$L^0_g := \sum_{i=1}^m L^0_{g_i}, \qquad \|g(x) - g(x')\| \le L^0_g \|x - x'\|.$$

Moreover, the Jacobian $J(x)$ is bounded for every $x \in \mathcal{X}$:

$$\|J(x)\| = \sqrt{\sum_{i=1}^{m} \|\nabla g_i(x)\|^2} \leq \sqrt{\sum_{i=1}^{m} (L_{g_i}^0)^2} \leq \sum_{i=1}^{m} L_{g_i}^0 = L_g^0.$$

(Definition of $\|A\|_F$ and its relation to other norms.)

The analysis below uses the tighter upper bound

$$\|J(x)\| \leq \sqrt{\sum_{i=1}^{m} (L_{g_i}^0)^2} =: M_g^0.$$

Optimality conditions of the algorithm updates

1. Optimality condition for the x update:

$$x^{k+1} = x^k - \tau \left(\nabla f(x^k) + (\nabla J(x^k))^T y^k \right).$$

2. Extrapolation on x:

$$\widetilde{x}^{k+1} = 2x^{k+1} - x^k, \qquad \widetilde{x}^k = 2x^k - x^{k-1}.$$

3. Optimality condition for the y update:

$$g^{k+1} = -M_k \left(y^{k+1} - y^k \right) + g(\widetilde{x}^{k+1}), \qquad g^{k+1} \in \partial h^*(y^{k+1}).$$

2 Convergence Analysis

Lemma 1 (dual boundedness)

Proof.

$$\begin{aligned}
\frac{\delta}{2}\|J(x^k)^T(y^{k+1} - y^k)\|^2 &= \frac{\delta}{2}\|(J(x^k) - J(x^{k+1}))^T y^{k+1} + J(x^{k+1})^T y^{k+1} - J(x^k)^T y^k\|^2 \\
&\leq \delta\|(y^{k+1})^T(J(x^{k+1}) - J(x^k))\|^2 + \delta\|J(x^{k+1})^T y^{k+1} - J(x^k)^T y^k\|^2 \\
&\leq \delta \sum_{i=1}^{m} \|y^{k+1}\|^2 \|\nabla g_i(x^{k+1}) - \nabla g_i(x^k)\|^2 \\
&\quad + \delta \left\| -\nabla f(x^{k+1}) + \frac{x^{k+1} - x^{k+2}}{\tau} + \nabla f(x^k) + \frac{x^{k+1} - x^k}{\tau} \right\|^2 \\
&\quad \left(x^{k+2} - x^{k+1} = -\tau \left(\nabla f(x^{k+1}) + J(x^{k+1})^T y^{k+1} \right) \right) \\
&\leq \delta M^2 \left(\sum_{i=1}^{m} L_{g_i}^2 \right) \|x^{k+1} - x^k\|^2 + 2\delta \left(\frac{1}{\tau} + L_f \right)^2 \|x^{k+1} - x^k\|^2 + \frac{2\delta}{\tau^2}\|x^{k+2} - x^{k+1}\|^2 \\
&\leq \left(\delta M^2 \sum_{i=1}^{m} L_{g_i}^2 + 2\delta \left(\frac{1}{\tau} + L_f \right)^2 \right) \|x^{k+1} - x^k\|^2 \\
&\quad + \frac{2\delta}{\tau^2}\|x^{k+2} - x^{k+1}\|^2. \qquad (2.1)
\end{aligned}$$

Lemma 2 (sufficient descent)

Proof. First, by the iteration formula of the algorithm with extrapolation on the primal variable and the convexity of h^*,

$$L(x^{k+1},y^{k+1})-L(x^k,y^k)\le-\left(\frac{1}{\tau}-\frac{L_f}{2}\right)\|x^{k+1}-x^k\|^2 + \langle y^{k+1},g(x^{k+1})\rangle-\langle y^k,g^k(x^{k+1})\rangle+\langle g^k,y^k-y^{k+1}\rangle. \tag{2.2}$$

For the second term on the right-hand side of (2.2), combine it with the last term by introducing $y^{k+1}-y^k$. Equation (2.2) is equivalently written as

$$L(x^{k+1},y^{k+1})-L(x^k,y^k)\le-\left(\frac{1}{\tau}-\frac{L}{2}\right)\|x^{k+1}-x^k\|^2 + \langle y^{k+1}-y^k,g(x^{k+1})-g^k\rangle+\langle y^k,g(x^{k+1})-g^k(x^{k+1})\rangle. \tag{2.3}$$

For the second term on the right-hand side of (2.3), using the optimality condition of the y^k update,

$$0\in\partial h^*(y^k)-g^k(\widetilde{x}^k)+\frac{1}{\sigma}(y^k-y^{k-1}),\qquad g^k\in\partial h^*(y^k),$$

we have

$$\begin{aligned}\langle y^{k+1}-y^k,g(x^{k+1})-g^k\rangle&=\langle y^{k+1}-y^k,g(x^{k+1})-g^k(\widetilde{x}^k)\rangle+\frac{1}{\sigma}\langle y^{k+1}-y^k,y^k-y^{k-1}\rangle\\&=\langle y^{k+1}-y^k,g(x^{k+1})-g^k(x^{k+1})\rangle\\&\quad+\left\langle y^{k+1}-y^k,g^k(x^{k+1})-g^k(\widetilde{x}^k)+\frac{1}{\sigma}(y^k-y^{k-1})\right\rangle.\end{aligned} \tag{2.4}$$

Substituting (2.4) into (2.3) gives

$$\begin{aligned}L(x^{k+1},y^{k+1})-L(x^k,y^k)\le&-\left(\frac{1}{\tau}-\frac{L}{2}\right)\|x^{k+1}-x^k\|^2+\langle y^{k+1},g(x^{k+1})-g^k(x^{k+1})\rangle\\&+\left\langle y^{k+1}-y^k,g^k(x^{k+1})-g^k(\widetilde{x}^k)+\frac{1}{\sigma}(y^k-y^{k-1})\right\rangle.\end{aligned} \tag{2.5}$$

We now bound the terms in (2.5).

For the second term on the right-hand side of (2.5),

$$\begin{aligned}\langle y^{k+1},g(x^{k+1})-g^k(x^{k+1})\rangle&\le\|y^{k+1}\|\,\|g(x^{k+1})-g^k(x^{k+1})\|\\&\le M\|g(x^{k+1})-g^k(x^{k+1})\|\qquad(\|y^k\|\le M)\\&\le M\sum_{i=1}^m\left|g_i(x^{k+1})-g_i^k(x^{k+1})\right|\\&\le\frac{ML_g^0}{2}\|x^k-x^{k+1}\|^2.\end{aligned} \tag{2.6}$$

Using the iteration formula, the last term on the right-hand side of (2.5) can be equivalently transformed as follows:

$$\left\langle y^{k+1}-y^k,g^k(x^{k+1})-g^k(\widetilde{x}^k)+\frac{1}{\sigma}(y^k-y^{k-1})\right\rangle$$

$$
\begin{aligned}
&= \left\langle y^{k+1}-y^k, J(x^k)\left[(x^{k+1}-x^k)-(x^k-x^{k-1})\right]+\frac{1}{\sigma}(y^k-y^{k-1})\right\rangle \\
&\qquad \left(g^k(z)=g(x^k)+J(x^k)(z-x^k), \quad \widetilde{x}^k=2x^k-x^{k-1}\right) \\
&= \left\langle y^{k+1}-y^k, \tau J(x^k)\left(\nabla f(x^{k-1})-\nabla f(x^k)+J(x^{k-1})^T y^{k-1}-J(x^k)^T y^k\right)+\frac{1}{\sigma}(y^k-y^{k-1})\right\rangle \\
&\qquad \left(x^k-x^{k-1}=-\tau\left(\nabla f(x^{k-1})+J(x^{k-1})^T y^{k-1}\right)\right) \\
&= \left\langle J(x^k)^T(y^{k+1}-y^k), \nabla f(x^{k-1})-\nabla f(x^k)\right\rangle \\
&\quad + \left\langle y^{k+1}-y^k, \left(\frac{1}{\sigma}I-\tau J(x^k)J(x^k)^T\right)(y^k-y^{k-1})\right\rangle \\
&\quad + \left\langle y^{k+1}-y^k, \tau J(x^k)\left(J(x^{k-1})-J(x^k)\right)^T y^{k-1}\right\rangle. \qquad (2.7)
\end{aligned}
$$

For the middle term in (2.7), if we set $\frac{1}{\sigma}I=\frac{1}{\sigma_k}I=M_k$, then this term vanishes. We continue to bound (2.7). Since we do not want y^k and y^{k+1} to appear, we combine the remaining two terms in (2.7) again:

$$
\begin{aligned}
&\left\langle y^{k+1}-y^k, g^k(x^{k+1})-g^k(\widetilde{x}^k)+M_k(y^k-y^{k-1})\right\rangle \\
&= \left\langle J(x^k)^T(y^{k+1}-y^k), \nabla f(x^{k-1})-\nabla f(x^k)+\tau J(x^k)\left(J(x^{k-1})-J(x^k)\right)^T y^{k-1}\right\rangle \\
&\le \frac{\delta}{2}\|J(x^k)^T(y^{k+1}-y^k)\|^2+\frac{1}{2\delta}\left\|\nabla f(x^{k-1})-\nabla f(x^k)+\tau J(x^k)\left(J(x^{k-1})-J(x^k)\right)^T y^{k-1}\right\|^2 \\
&\qquad \left(\langle a,b\rangle\le\frac{\delta}{2}\|a\|^2+\frac{1}{2\delta}\|b\|^2\right). \qquad (2.8)
\end{aligned}
$$

For the second term in (2.8),

$$
\begin{aligned}
&\frac{1}{2\delta}\left\|\nabla f(x^{k-1})-\nabla f(x^k)+\tau J(x^k)\left(J(x^{k-1})-J(x^k)\right)^T y^{k-1}\right\|^2 \qquad (2.9) \\
&\le \frac{1}{\delta}\|\nabla f(x^k)-\nabla f(x^{k-1})\|^2+\frac{\tau^2}{\delta}\|J(x^k)\left(J(x^{k-1})-J(x^k)\right)^T y^{k-1}\|^2 \\
&\le \frac{L_f^2}{\delta}\|x^k-x^{k-1}\|^2+\frac{\tau^2}{\delta}\left\|\sum_{i=1}^m (y^{k-1})^T\left(J(x^{k-1})-J(x^k)\right)(J(x^k))^T\right\|^2 \\
&\le \frac{L_f^2}{\delta}\|x^k-x^{k-1}\|^2+\frac{\tau^2}{\delta}\sum_{i=1}^m\left\|(y^{k-1})^T(\nabla g_i(x^{k-1})-\nabla g_i(x^k))(\nabla g_i(x^k))^T\right\|^2 \\
&\le \frac{L_f^2}{\delta}\|x^k-x^{k-1}\|^2+\frac{\tau^2}{\delta}\sum_{i=1}^m\|y^{k-1}\|^2\|\nabla g_i(x^k)\|^2\|\nabla g_i(x^{k-1})-\nabla g_i(x^k)\|^2 \\
&\le \frac{L_f^2+\sum_{i=1}^m(\tau M L_{g_i}^0 L_{g_i})^2}{\delta}\|x^k-x^{k-1}\|^2. \qquad (2.10)
\end{aligned}
$$

For the first term in (2.8),

$$
\frac{\delta}{2}\|J(x^k)^T(y^{k+1}-y^k)\|^2=\frac{\delta}{2}\|(J(x^k)-J(x^{k+1}))^T y^{k+1}+J(x^{k+1})^T y^{k+1}-J(x^k)^T y^k\|^2
$$

$$
\begin{aligned}
&\leq \delta\|(y^{k+1})^T(J(x^{k+1})-J(x^k))\|^2+\delta\|J(x^{k+1})^Ty^{k+1}-J(x^k)^Ty^k\|^2\\
&\leq \delta\sum_{i=1}^m\|y^{k+1}\|^2\|\nabla g_i(x^{k+1})-\nabla g_i(x^k)\|^2\\
&\quad+\delta\left\|-\nabla f(x^{k+1})+\frac{x^{k+1}-x^{k+2}}{\tau}+\nabla f(x^k)+\frac{x^{k+1}-x^k}{\tau}\right\|^2\\
&\qquad\left(x^{k+2}-x^{k+1}=-\tau\left(\nabla f(x^{k+1})+J(x^{k+1})^Ty^{k+1}\right)\right)\\
&\leq \delta M^2\left(\sum_{i=1}^m L_{g_i}^2\right)\|x^{k+1}-x^k\|^2+2\delta\left(\frac{1}{\tau}+L_f\right)^2\|x^{k+1}-x^k\|^2+\frac{2\delta}{\tau^2}\|x^{k+2}-x^{k+1}\|^2\\
&\leq \left(\delta M^2\sum_{i=1}^m L_{g_i}^2+2\delta\left(\frac{1}{\tau}+L_f\right)^2\right)\|x^{k+1}-x^k\|^2+\frac{2\delta}{\tau^2}\|x^{k+2}-x^{k+1}\|^2.
\end{aligned}
\tag{2.11}
$$

Combining the above inequalities, we obtain

$$
\begin{aligned}
L(x^{k+1},y^{k+1})-L(x^k,y^k)\leq&-\left(\frac{1}{\tau}-\frac{L}{2}\right)\|x^{k+1}-x^k\|^2+\langle y^{k+1}-y^k,g(x^{k+1})-g(\widetilde{x}^k)\rangle\\
&+\frac{1}{\sigma}\langle y^{k+1}-y^k,y^k-y^{k-1}\rangle\\
&+\langle y^k,g(x^{k+1})-g^k(x^{k+1})\rangle\\
&\left(\text{convexity of } h^*\text{, the } y^k \text{ update, and the } x^{k+1} \text{ update}\right),
\end{aligned}
$$

where

$$
g(\widetilde{x}^k)=g(2x^k-x^{k-1}),\qquad g^k(x^{k+1})=g(x^k)+J(x^k)(x^{k+1}-x^k).
$$

For the second term on the left-hand side of the inequality,

$$
\begin{aligned}
\langle y^{k+1}-y^k,g(x^{k+1})-g(\widetilde{x}^k)\rangle\leq&\frac{\delta}{2}\|y^{k+1}-y^k\|^2+\frac{1}{2\delta}\|g(x^{k+1})-g(\widetilde{x}^k)\|^2\\
\leq&\frac{\delta}{2}\|y^{k+1}-y^k\|^2+\frac{L}{2\delta}\|x^{k+1}-x^k+x^{k-1}-x^k\|^2\\
&(g_i(u)\text{ is }L_{g_i}\text{-Lipschitz continuous})\\
\leq&\frac{\delta}{2}\|y^{k+1}-y^k\|^2+\frac{L\tau^2}{2\delta}\|\nabla f(x^{k-1})-\nabla f(x^k)+J(x^{k-1})^Ty^{k-1}-J(x^k)^Ty^k\|^2\\
\leq&\frac{\delta}{2}\|y^{k+1}-y^k\|^2+\frac{LL_f^2\tau^2}{\delta}\|x^{k-1}-x^k\|^2+\frac{L\tau^2}{\delta}\|J(x^{k-1})^Ty^{k-1}-J(x^k)^Ty^k\|^2\\
&(f(x)\text{ is }L_f\text{-smooth}).
\end{aligned}
$$

Here, $x^{k+1}=x^k-\tau(\nabla f(x^k)+J(x^k)^Ty^k)$, using the x^k update.

For the red part above, can we impose an assumption on the Jacobian analogous to the linear case (Ax), so that this part can be bounded by $c\|y^k-y^{k-1}\|^2$?

For the fourth term on the left-hand side of the inequality, combining the above discussion with the one-side accuracy assumption from the weakly convex paper discussed earlier,

$$
\begin{aligned}
\langle y^k,g(x^{k+1})-g^k(x^{k+1})\rangle&\leq M\|g(x^{k+1})-g^k(x^{k+1})\|\qquad(\|y^k\|\leq M)\\
&\leq\frac{M\mu}{2}\|x^k-x^{k+1}\|^2
\end{aligned}
$$

Here, $g_i(u)$ is infinitely continuously differentiable and L-smooth under the bounded-sequence assumption; the Hessian can be bounded.

For the third term, it cannot be canceled as in the linear-constraint case. Consider the following bound:

$$\frac{1}{\sigma}\langle y^{k+1}-y^k, y^k-y^{k-1}\rangle \le \frac{1}{2\sigma}\left(\|y^k-y^{k+1}\|^2+\|y^k-y^{k-1}\|^2\right).$$

$$L(x,y) = f(x) + \langle y, g(x)\rangle - h^*(y). \tag{2.12}$$

Lemma 3 (subgradient boundedness)

Define

$$\kappa_1 = \frac{14}{\tau^2},$$

$$\kappa_2 = 3\left(L_f^2 + \frac{1}{\tau^2} + 4M^2\sum_{i=1}^m L_{g_i}^2 + 4\left(\frac{1}{\tau}+L_f\right)^2\right),$$

$$\kappa_3 = 2(M_g^0)^2\left(1+\tau^2\left(M^2\sum_{i=1}^m L_{g_i}^2 + 2\left(\frac{1}{\tau}+L_f\right)^2\right)\right), \qquad \kappa = \max\{\kappa_1, \kappa_2+\kappa_3\},$$

and $u^k = \{x^k, x^{k-1}\}$. Then,

$$\operatorname{dist}(0, \partial L(x^k, y^k)) \le \kappa_1\|x^{k+1}-x^k\|^2 + (\kappa_2+\kappa_3)\|x^k-x^{k-1}\|^2 \tag{2.13}$$

$$\le \kappa\|u^{k+1}-u^k\|^2. \tag{2.14}$$

$$0 \in \partial L(x^k, y^k),$$

$$d_x^k = \nabla_x L(x^k) + J(x^k)^T y^k, \qquad \partial L(x^k) \ni d_y^k = g(x^k) - g_y^k, \quad \text{where } g^k \in \partial h^*(y^k).$$

For the first part, according to the optimality condition of the x^k update,

$$\|d_1^k\|^2 = \|(\nabla f(x^k) - \nabla f(x^{k-1})) + (\nabla f(x^{k-1}) + J(x^{k-1})^T y^{k-1}) + (J(x^k)^T y^k - J(x^{k-1})^T y^{k-1})\|^2 \tag{2.15}$$

$$\le 3\left(\|\nabla f(x^k) - \nabla f(x^{k-1})\|^2 + \|\nabla f(x^{k-1}) + J(x^{k-1})^T y^{k-1}\|^2 + \|J(x^k)^T y^k - J(x^{k-1})^T y^{k-1}\|^2\right)$$

$$\le 3\left(L_f^2 + \frac{1}{\tau^2}\right)\|x^k - x^{k-1}\|^2 + 3\|J(x^k)^T y^k - J(x^{k-1})^T y^k + J(x^{k-1})^T y^k - J(x^{k-1})^T y^{k-1}\|^2$$

$$\le 3\left(L_f^2 + \frac{1}{\tau^2}\right)\|x^k - x^{k-1}\|^2 + 6\|(J(x^k) - J(x^{k-1}))^T y^k\|^2 + 6\|J(x^{k-1})^T(y^k - y^{k-1})\|^2$$

$$\le 3\left(L_f^2 + \frac{1}{\tau^2} + 2M^2\sum_{i=1}^m L_{g_i}^2\right)\|x^k - x^{k-1}\|^2 \tag{2.16}$$

$$+ 6\left(M^2\sum_{i=1}^m L_{g_i}^2 + 2\left(\frac{1}{\tau}+L_f\right)^2\right)\|x^k - x^{k-1}\|^2 + \frac{12}{\tau^2}\|x^{k+1} - x^k\|^2$$

$$= \frac{12}{\tau^2}\|x^{k+1} - x^k\|^2 + 3\left(L_f^2 + \frac{1}{\tau^2} + 4M^2\sum_{i=1}^m L_{g_i}^2 + 4\left(\frac{1}{\tau}+L_f\right)^2\right)\|x^k - x^{k-1}\|^2.$$

The inequality (2.16) follows from Lemma 1, the Lipschitz continuity of $J(x^k)$, and norm compatibility. The second part: according to y